

CS 418 - PROJECT PROPOSAL

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Project Name and Github:

Project Name: Climate Change-Induced Migration Prediction

Github: <https://github.com/sharadruthi-uic> (couldn't link my personal Github profile due to an authentication issue, will make sure to add it by the next submission)

This project aims to develop a predictive model that identifies migration patterns driven by climate change, focusing on how extreme weather events and resource scarcity influence displacement. The goal is to develop predictive models to aid humanitarian organizations and policymakers in planning and response efforts.

Problem:

The central question is: How can we predict climate-driven migration patterns from factors like extreme weather and sea-level rise? Climate change is escalating displacement, potentially affecting up to 216 million people by 2050, primarily through internal migration. Addressing this issue will help governments and organizations allocate resources effectively. I chose this problem due to its growing relevance and the role data science can play in creating actionable solutions. I hypothesize that regions facing severe environmental stressors will become migration hotspots, requiring targeted interventions. ([Migration data portal](#))

Data:

The project will utilize multiple data sources, including the UNHCR Population Statistics Database, World Bank Migration and Remittances data, IPCC climate projections, and NOAA climate data. These datasets include refugee and migration statistics, climate variables like temperature and precipitation, and socio-economic indicators. Data will be collected and preprocessed to create a unified dataset, and the size may vary based on temporal and spatial scale alignment, estimated to have millions of rows per dataset.

Solution:

I plan to approach the problem using machine learning models like Random Forests and time-series forecasting to predict migration patterns under different climate scenarios. I will start with exploratory data analysis (EDA) to uncover patterns between climate and migration variables. Using GIS tools and clustering techniques for spatial analysis, I'll map hotspots of migration. Agent-based modeling will simulate migration decisions. The end result will be an interactive dashboard that visualizes migration risk zones and policy simulations. For the progress report, I aim to have preliminary EDA completed and initial models built.